

Employee Attrition Analysis

Exploratory Data Analysis (EDA) Report

1. Introduction

Exploratory Data Analysis (EDA) was performed to understand employee characteristics and identify factors influencing employee attrition.

2. Univariate Analysis

The distribution of employee age, income, department, gender, job role, education, marital status, and years at company was analyzed.

Key Findings:

- Most employees belong to the age group of 30–40 years.
- Research & Development has the highest workforce.
- Male employees outnumber female employees.
- Most employees possess Bachelor's degrees.
- Employee tenure is concentrated in lower experience ranges.

3. Attrition Analysis

Employee attrition was analyzed across various employee segments.

Key Findings:

- Overall attrition rate is 16.12%.
- Sales department exhibits comparatively higher attrition.
- Sales Executives and Laboratory Technicians experience higher turnover.
- Employees who travel frequently show higher attrition.
- Employees working overtime exhibit significantly higher attrition.

4. Satisfaction and Workplace Factors

The impact of job satisfaction, work-life balance, and environment satisfaction on attrition was examined.

Key Findings:

- Lower job satisfaction is associated with higher attrition.
- Poor work-life balance contributes to employee turnover.
- Employees with low environment satisfaction are more likely to leave.

5. Salary and Tenure Analysis

Salary and employee tenure were analyzed against attrition.

Key Findings:

- Employees with lower monthly income are more likely to leave.
- Attrition is more prominent among employees with shorter tenure.

6. Correlation Analysis

A correlation heatmap was generated to identify relationships among numerical variables.

Key Findings:

- Monthly Income and Job Level show strong positive correlation.
- Years at Company is positively related to Years in Current Role.
- Total Working Years is positively associated with Monthly Income.

7. Conclusion

The analysis revealed that overtime, job satisfaction, work-life balance, business travel, salary, and employee tenure are major factors influencing employee attrition.